



Frozen fish

冰封之鱼

难度评级：★★★★☆

How salmon evolved to cope with the ice age

鲑鱼如何演化以适应冰河时代

【 Para.1 】

Some species thrived during the last ice age. Sabre-toothed cats, giant ground sloths, woolly mammoths and woolly rhinos all did pretty well. Others, however, were seriously inconvenienced—among them the chum salmon of north-western North America, which frequently found the rivers leading to and from their inland **spawning** grounds

icebound.

部分物种在上个冰河时代繁衍兴盛。剑齿虎、大地懒、猛犸象以及披毛犀都活得相当顺遂。然而另有一些物种却深陷生存困境，北美西北部的大马哈鱼就是其中之一：它们往返内陆产卵地的河道屡屡被冰层封冻。

【 Para.2 】

文章导读：

冰河时代部分物种繁盛，北美狗鲑因河道冰封遭遇生存危机。研究发现部分被困狗鲑并未灭绝，反而繁衍兴盛。太平洋狗鲑部分种群存在特殊基因与行为变异，令研究者费解。马丁猜想：冰期隔离于冰川湖造成狗鲑基因分化。团队采集大量样本开展基因分析验证猜想。排除亚洲鲑鱼基因渗入，证实种群在冰川湖隔离演化。被困狗鲑演化出新行为，冰坝消融后部分习性得以留存。解读特殊基因来源，该研究对科研、物种保护、水产养殖均有意义。

spawn v. /spɔːn/ (鱼、蛙等) 产卵；引发

SYN: lay eggs, generate

Rapid digital transformation **spawns** numerous new-form occupations that barely existed decades ago.

飞速的数字化转型催生了许多数十年前几乎不存在的新型职业。

icebound adj. /'aɪsbəʊnd/ 被冰封住的；封冻的

SYN: ice-covered

During harsh winters, several northern transport channels become **icebound** and restrict regional trade activities.

严寒冬季，北方数条运输航道被封冻，制约区域贸易活动。

nuisance n. /'njuːns/ 麻烦事；令人讨厌的事物 【考研大纲词汇】

SYN: trouble, annoyance

Random noise in residential communities has become a constant **nuisance** affecting residents' quality of life.

居民区的随意噪声已经成为持续困扰居民生活质量的麻烦事。

alternative adj. /ɔːl'tɜːnətɪv/ 可供替代的；另外的 【考研大纲词汇】

SYN: substitute, optional

It is wise for the whole society to develop **alternative** energy resources to reduce reliance on fossil fuels.

全社会开发替代能源，降低对化石燃料的依赖，是明智之举。

accessible adj. /ək'sesəbl/ 可到达的；可获取的；易进入的 【考研大纲词汇】

SYN: reachable, available

Public cultural resources ought to be made **accessible** to citizens from all age groups and social classes.

公共文化资源应当保证各年龄层、各社会阶层公民都能够获取。

marine adj. /mə'riːn/ 海洋的；海生的 【考研大纲词汇】

SYN: oceanic

The protection of **marine** ecological environment has become a key topic within global environmental governance.

海洋生态环境保护已经成为全球环境治理中的核心议题。

For those cut off in the sea, this was a **nuisance**. They had to make **alternative** arrangements, by seeking other, more **accessible** bodies of fresh water in which to lay their eggs. Those cut off on the land side, though, were stuck. And it was assumed until recently that because they were unable to complete the **marine** part of their life-cycle, they died out.

对于被困在海洋中的鲑鱼而言，这只是个麻烦。它们不得不另寻出路，去找其他更容易抵达的淡水水域产卵。但被困在陆地一侧的鲑鱼则陷入绝境。直到不久前，人们还认为，由于它们无法完成生命周期中的海洋阶段，最终走向了灭绝。

【 Para.3 】

Not so, it turns out. A study published recently in Quaternary Research by Patrick Martin, an **avid** fisherman with a **sideline** as an **amateur** scientist, has employed genetics to show that some of the left-behinds not only **endured**, but thrived.

然而事实并非如此。帕特里克·马丁既是狂热的垂钓爱好者，同时也是一名业余科学家，他近期在《第四纪研究》期刊上发表的一项研究借助遗传学手段证实，部分被困滞留的鲑鱼非但存活了下来，甚至还繁衍兴盛。

【 Para.4 】

Pacific chum salmon range around that ocean's **rim** from Japan and the Koreas to the coast of Oregon. Naturally, there is genetic variation in such a widespread population. Some of this variation is, though, so extreme that it has left

avid adj. /'ævid/ 热衷的；酷爱；渴求的

SYN: enthusiastic, keen

An **avid** natural-history observer can accumulate abundant first-hand materials during long-term field trips.

一位热衷于博物学的观察者可以在长期野外考察中积累大量一手资料。

sideline n. /'saɪdlɪn/ 副业；兼职；附带活动 【考研大纲词汇】

SYN: side job

For many young people, creative content creation serves as a rewarding **sideline** apart from full-time occupations.

对于许多年轻人而言，创作内容是全职工作之外一份收益可观的副业。

amateur adj. /'æmətʃə(r)/ 业余的；非专业的 【考研大纲词汇】

SYN: non-professional

Some valuable discoveries in natural science are contributed by **amateur** researchers rather than professional experts.

自然科学领域部分有价值的发现来自业余研究者，而非专业专家。

endure v. /ɪn'dʒʊə(r)/ 忍受；熬过；存活 【考研大纲词汇】

SYN: survive, withstand

Those who **endure** setbacks and frustrations in life are more likely to obtain lasting achievements.

那些在人生中熬过挫折磨难的人，更有可能取得长久的成就。

rim n. /rɪm/ 边缘；周边地带 【考研大纲词汇】

SYN: edge, margin

Many new-built economic zones are distributed along the **rim** of big cities to ease the pressure of central districts. 不少新建经济区分布在大城市周边，用以缓解中心城区压力。

baffled adj. /'bæfld/ 困惑的；难以理解的 【考研大纲词汇】

SYN: puzzled, confused

Scholars remain **baffled** by why some well-intentioned policies fail to achieve expected practical outcomes.

为何部分初衷良好的政策无法取得预期实效，依旧令学者感到困惑。

intriguingly adv. /ɪn'tri:gɪŋli/ 耐人寻味地；引人好奇地

SYN: curiously

Intriguingly, higher material wealth does not always bring about corresponding growth in personal happiness.

耐人寻味的是，物质财富的增加并不总能带来个人幸福感的同步提升。

migrate v. /maɪ'greɪt/ 迁徙；（鱼类）洄游；迁移 【考研大纲词汇】

SYN: move, travel

Large numbers of young labourers **migrate** from underdeveloped regions to big cities in pursuit of better prospects.

大量年轻劳动力从不发达地区迁往大城市，追求更好的发展前景。

speculate v. /'spekjuleɪt/ 推测，猜想；思索 【考研大纲词汇】

SYN: conjecture, hypothesize

Economists often **speculate** about future consumption trends according to current household-spending data.

经济学家常常依据当下家庭消费数据推测未来消费趋势。

researchers **baffled**. Three particular hot spots are around Kodiak island and nearby Cook inlet, in Alaska, and in the Salish sea between British Columbia and Washington state.

太平洋狗鲑分布于太平洋沿岸，从日本、朝鲜半岛一直延伸至俄勒冈州海岸。这类分布如此广泛的种群，内部自然存在遗传变异。不过，其中部分变异程度极大，令研究人员困惑不解。三处特定的热点区域分别位于阿拉斯加的科迪亚克岛及其附近的库克湾，以及不列颠哥伦比亚省与华盛顿州之间的萨利希海。

【 Para.5 】

Intriguingly, it is not only the fishes' genetics that differ, but also their behaviours. Most salmon **migrate** from the sea into fresh water to spawn in autumn or winter. The Salish sea population does so in summer. Similarly, salmon generally spawn in rivers. One of the Cook inlet populations, by contrast, does so in a lake.

耐人寻味的是，发生差异的不仅是这些鲑鱼的基因，还有它们的行为习性。大多数鲑鱼会在秋季或冬季从海洋洄游至淡水水域产卵。而萨利希海种群却选择在夏季产卵。同样，鲑鱼一般都在河流中产卵。与之形成对比的是，库克湾的某一个种群却在湖泊里产卵。

【 Para.6 】

Isolation over periods of thousands of years is well understood to drive evolution. This is obvious on small islands, such as the Galápagos. But walls of ice can isolate, too. With this in mind, Mr Martin **speculated** that the Kodiak, Cook and Salish salmon are genetically different from their neighbours because they got trapped, and

hunch n. /hʌntʃ/ 直觉；预感，猜想
SYN: intuition, guess
One should not make crucial policy decisions merely based on a **hunch**; sufficient data support is indispensable.
人们不应当仅凭猜想做出关键决策，充分的数据支撑必不可少。

convene v. /kən'vi:n/ 召集；组建（团队）；召开 【考研大纲词汇】
SYN: assemble, organize
The government will **convene** an expert panel to work out practical solutions for urban traffic congestion.
政府将召集专家小组，制定切实可行的城市拥堵解决方案。

specimen n. /'spesɪmən/ 标本；样本，样品 【考研大纲词汇】
SYN: sample
Researchers collect mass questionnaire **specimens** to analyse young people's attitudes toward social mobility.
研究者收集大量问卷样本，分析年轻人对于社会流动的态度。

aberrant adj. /æ'berənt/ 反常的；偏离常规的；变异的
SYN: abnormal, deviant
Moral education is supposed to correct **aberrant** social behaviours and guide citizens to abide by public norms.
道德教育旨在纠正反常的社会行为，引导公民遵守公共准则。

bear no close relation to phrase 与.....无密切亲缘/关联
SYN: have little kinship with
Numerous studies indicate that high scores **bear no close relation to** one's comprehensive practical competence.
大量研究表明，高分与一个人的综合实践能力并无密切关联。

kin n. /kɪn/ 亲属；同类，同族生物 【考研大纲词汇】
SYN: relatives, kindred
Traditional ethics attach great importance to maintaining emotional bonds with one's **kin** and community members.
传统伦理十分重视维系与亲属及社群成员之间的情感纽带。

rejig v. /'ri:'dʒɪg/ 重新调整；改造，改写
SYN: readjust, remodel
Many universities **rejig** their talent-training schemes to satisfy the changing demands of modern industries.
许多高校重新调整人才培养方案，适配现代产业不断变化的需求。

miniature adj. /'mɪnɪətʃə(r)/ 微型的；小规模 【考研大纲词汇】
SYN: tiny, small-scale
A well-run community can be regarded as a **miniature** society reflecting various realistic social contradictions.
运转良好的社区可以视作一个微型社会，折射出各类现实社会矛盾。

liberate v. /'lɪbəreɪt/ 释放；使摆脱束缚 【考研大纲词汇】
SYN: free, release
Popular digital tools **liberate** ordinary people from complicated manual work and greatly improve work efficiency.
普及的数字工具把普通人从繁杂手工劳动中解放出来，大幅提升工作效率。

survived, in glacial lakes.

数千年来的地理隔绝会推动生物进化，这已成为共识。在加拉帕戈斯这类小型岛屿上，这一现象尤为明显。不过，冰墙同样也能造成地理隔离。受此启发，马丁先生推测，科迪亚克、库克湾以及萨利希海的鲑鱼之所以和邻近种群存在基因差异，是因为它们当年被困在冰川湖泊之中，并且存活了下来。

【 Para.7 】

Between 23,000 and 15,000 years ago glaciers created a lake on Kodiak island, with an area of 1,000 square kilometres and a depth of 300 metres. Glacial lakes also formed in valleys around the Olympic mountains of modern-day Washington state. To test his **hunch**, Mr Martin therefore **convened** a team of ichthyologists and geologists from the University of Washington and America's National Oceanic and Atmospheric Administration to analyse the genes of 32,817 **specimens** from 310 populations around the Pacific rim.

在2.3万年至1.5万年前，冰川在科迪亚克岛上形成了一座面积达1000平方公里、水深300米的湖泊。同样，冰川湖也出现在如今华盛顿州奥林匹克山脉周边的山谷之中。为验证自己的猜想，马丁组建了一支由华盛顿大学及美国国家海洋和大气管理局的鱼类学家与地质学家组成的研究团队，对太平洋沿岸310个鲑鱼种群共计32817份样本开展基因分析。

【 Para.8 】

One question that needed an answer was whether the unusual genetics could be explained by a few salmon from

peculiarity n. /prɪˈkjuːliːərəti/ 独特性；特质；奇特之处
【考研大纲词汇】

SYN: oddity, distinct feature

Cultural **peculiarity** constitutes an essential source of soft power for a nation amid global integration.

在全球一体化进程中，文化独特性是一国软实力的重要来源。

presumably adv. /prɪˈzjuːməbli/ 大概；据推测；想必
【考研大纲词汇】

SYN: supposedly, probably

Presumably, excessive pursuit of short-term economic returns will exert negative impacts on long-term social development.

据推测，过度追求短期经济回报会对社会长远发展产生负面影响。

relic n. /ˈreɪlɪk/ 遗迹；遗留物；残存特征 【考研大纲词汇】

SYN: vestige, remnant

Some deep-rooted social prejudices are **relics** of outdated value systems that have lingered for centuries.

部分根深蒂固的社会偏见是陈旧价值体系遗留下来的产物，已经存续数百年。

physiological adj. /ˌfɪziəˈlɒdʒɪkl/ 生理的；生理学的 【考研大纲词汇】

SYN: bodily

Long-term mental pressure may trigger various **physiological** disorders and weaken individuals' immunity.

长期精神压力会引发各类生理紊乱，削弱个体免疫力。

briny n. /'brɪni/ n. 海洋，海水

SYN: salt water

Coastal wetlands serve as a buffer zone between the **briny** and inland terrestrial ecosystems.

滨海湿地是海洋咸水环境与内陆陆地生态系统之间的缓冲地带。

insurance n. /ɪnˈʃʊərəns/ 保险；保障，防护手段 【考研大纲词汇】

SYN: safeguard, protection

Diversified skill sets act as reliable **insurance** against unemployment in an increasingly competitive job market.

在竞争愈发激烈的就业市场，多元化技能是抵御失业风险的可靠保障。

extinction n. /ɪkˈstɪŋkʃn/ 灭绝；消亡 【考研大纲词汇】

SYN: extermination, disappearance

Blind exploitation of natural resources may accelerate the extinction of numerous rare wildlife species.

对自然资源的盲目开发会加速大量珍稀野生物种的消亡。

in spades idiom 大量地；非常，极度

SYN: in abundance, to a great degree

Metropolises possess job opportunities and educational resources **in spades** yet face severe urban problems simultaneously.

大都市拥有海量就业与教育资源，但同时也面临严峻的城市问题。

pathogens n. /ˈpæθədʒən/ 病原体，致病菌

SYN: disease-causing agent

Improved public sanitation facilities can effectively block the spread of harmful **pathogens** in crowded communities.

完善的公共卫生设施能够有效阻断有害病原体在人口密集社区传播。

Asia having taken a wrong turn and introduced their DNA into their North American cousins. The **aberrant** populations, however, **bore no close relation to** their Asian **kin**. Instead, the team proposes that the ancestors of these survived the ice age isolated in freshwater lakes, where they went their own sweet evolutionary ways.

有一个亟待解答的问题：这些不同寻常的基因特征，是否源于少数来自亚洲的鲑鱼误入歧途，将自身DNA带入了北美同类种群之中。然而，这些变异种群和亚洲同类并无密切亲缘关系。研究团队由此提出假说：这些鲑鱼的祖先在冰河时代被困在淡水湖泊中与世隔绝，并在此走上了独属于自己的进化道路，最终存活下来。

【 Para.9 】

During this period, Mr Martin and his colleagues argue, the fish **rejigged** their behaviour to treat the lakes they were trapped in like **miniature** oceans, complete with migrating up glacial streams feeding the lakes to spawn. That rejigging then persisted, at least in part, when the ice dams melted and the fish were **liberated** into the sea.

马丁及其同事提出，在这一时期，这些鱼类调整了自身行为，把困住它们的湖泊视作微型海洋，并会洄游到汇入湖泊的冰川溪流之中产卵。之后冰坝消融，这些鲑鱼得以重返海洋，但这种行为上的调整至少有一部分保留了下来。

【 Para.10 】

At least some of chum salmon's genetic **peculiarities** are

今日句子分析

主句

A study published recently in Quaternary Research by

主语

后置定语 1

Patrick Martin, an avid fisherman with a sideline as an

后置定语 2

同位语

amateur scientist, has employed genetics to show

谓语

宾语

状语

宾语从句

that some of the left-behinds not only endured, but

从句主语

并列谓语 1

thrived.

并列谓语2

句子分析：

(1) 主句主干

主语：A study；核心名词study“研究”。

后置定语1：published recently in Quaternary Research，过去分词短语作后置定语修饰study；recently为时间副词；in Quaternary Research介词短语，说明发表期刊。

后置定语2：by Patrick Martin，介词by引出该研究的作者。

同位语：an avid fisherman with a sideline as an amateur scientist，逗号隔开作Patrick Martin的同位语；with a sideline as an amateur scientist介词短语修饰fisherman，交代人物双重身份：垂钓爱好者，副业是业余科学家。

谓语：has employed，现在完成时；employ此处熟词僻义，意为“运用、采用”。

宾语：genetics，名词，“遗传学（手段）”。

目的状语：to show，不定式，说明运用遗传学的目的。

(2) show的宾语从句

引导词that，不可省略，引导宾语从句作show的宾语。

从句主语：some of the left-behinds；left-behinds合成名词，文中指冰河时代被困住的鲑鱼。

并列谓语：not only endured, but thrived；not only...but (but)...并列结构，连接两个不及物动词。endure表示存活、熬过困境；thrive表示繁衍兴盛。

(3) 翻译重难点

① employ为熟词僻义，不要译作“雇佣”，科研语境译为“借助、运用”。

② left-behinds是文章临时合成名词，词典无收录，结合上下文不能直译“落在后面的东西”，指代前文被冰封河道困住的鲑鱼，译为“部分被困滞留的鲑鱼”。

③ not only endured, but thrived存在语义递进：先是艰难存活，进而繁衍兴盛，翻译要体现递进逻辑，不要简单译为“不仅存活而且活得好”。

thus, **presumably**, the drivers of their unusual behaviours. But others could be **relics** of **physiological** adaptations to the stress of a life confined to fresh water in a species that normally spends much of its existence in the **briny**. That is of interest to evolutionary biologists. It may also intrigue conservationists, for genetic diversity is a good **insurance** against **extinction** and these odd salmon populations have diversity **in spades**. And fish farmers may take notice as well, if they are thinking about trying to raise salmon in colder waters or even freshwater lakes, where sea lice are absent and infectious **pathogens** less common.

因此，狗鲑的部分基因特异性正是催生其特殊行为的驱动因素。但另一些基因特征则可能是生理适应留下的遗迹：该物种原本大部分时间都生活在咸水环境，却被迫长期困在淡水之中承受生存压力，由此产生了生理适应。这一点对进化生物学家而言颇具研究价值。这同样会引起保护主义者的兴趣，因为遗传多样性是抵御物种灭绝的有力保障，而这些特殊的鲑鱼种群拥有极高的多样性。水产养殖户或许也会对此加以关注，如果他们打算在更寒冷的水域、甚至淡水湖泊养殖鲑鱼——这类水域没有海虱，传染性病原体也更为稀少。

④ 长同位语处理：中文习惯把人物身份提前，不要生硬后置。

整句翻译：帕特里克·马丁既是狂热的垂钓爱好者，同时也是一名业余科学家，他近期在《第四纪研究》期刊上发表的一项研究借助遗传学手段证实，部分被困滞留的鲑鱼非但存活了下来，甚至还繁衍兴盛。

➤ 写作积累

(Para. 1) Some species thrived during the last ice age. Sabre-toothed cats, giant ground sloths, woolly mammoths and woolly rhinos all did pretty well. Others, however, were seriously inconvenienced—among them the chum salmon of north-western North America, which frequently found the rivers leading to and from their inland spawning grounds icebound.

本段一共三句话，前两句交代时代背景，介绍冰河时代部分物种得以蓬勃繁衍，并列剑齿虎、大地懒等史前动物作为实例。第三句借助however完成语义转折，转到处境艰难的另一批物种；破折号引出本文主角——北美西北部的狗鲑，随后用which引导非限制性定语从句对狗鲑展开说明。从句里leading to and from their inland spawning grounds是现在分词短语，作后置定语修饰rivers；find the rivers...icebound属于find+宾语+宾语补足语结构，表达“发现河道处于冰封状态”。

本段出现一组冰河时代古生物名词：Sabre-toothed cats (/ˌseɪbəˈtuːθt kæts/)，剑齿虎，史前大型食肉猫科，长有修长锋利的犬齿；giant ground sloths (/ˈdʒaɪənt graʊnd sləʊs/)，大地懒，栖息于美洲的巨型陆生树懒，现已灭绝；woolly mammoths (/ˈwʊliˈmæməʊs/)，长毛猛犸象，身披厚毛，适应冰河严寒，已经灭绝；woolly rhinos (/ˈwʊliˈraɪnəʊz/)，披毛犀，冰河时期耐寒大型犀牛，同样已经灭绝；chum salmon (/tʃʌmˈsæmən/)，狗鲑（大马哈鱼），太平洋鲑鱼的一种，也是文章的核心研究对象。

前四个都是更新世已灭绝的史前动物，只有狗鲑是现存物种。重点词汇方面，thrive可以描述物种繁衍兴盛，也可以迁移到生态、社会话题写作；spawning grounds是科普专有表达，指鱼类的产卵地；icebound为形容词，意思是被冰层封冻。写作逻辑上，文段先对比两类生物截然不同的生存状态，列举代表性物种，从普遍现象落到具体研究个案；破折号用来切换叙述重心，定语从句补充细节，是科普文很实用的写法。仿写时可以借鉴这套思路：先摆出两极分化的客观现象，再把视角收拢到某一个具体对象上细化阐述。

(Para. 8) One question that needed an answer was whether the unusual genetics could be explained by a few salmon from Asia having taken a wrong turn and introduced their DNA into their North American cousins.

这句话抛出研究要厘清的核心疑问，整体为主系表结构，主语One question后接that引导的定语从句修饰question，表语是whether引导的表语从句，用来呈现待验证的猜想。表语从句内部a few salmon from Asia having taken a wrong turn and introduced their DNA into their North American cousins是动名词复合结构，作介词by的宾语，用来解释基因异常的一种可能性假设。

(Para. 10) It may also intrigue conservationists, for genetic diversity is a good insurance against extinction and these odd salmon populations have diversity in spades.

这句话属于并列复合句。主句是 It may also intrigue conservationists, it指代前文鲑鱼特殊演化这一研究发现；for在这里是并列连词，引导原因分句，并非介词，用来补充解释为什么会引起环保人士关注，for后面由and连接两个并列分句：genetic diversity is a good insurance against extinction和these odd salmon populations have diversity in spades，一讲理论原理，二讲本文鲑鱼种群的实际情况。

intrigue此处是动词，意思是“激起……的兴趣、引发好奇”，不是名词“阴谋”，主语常是研究、现象、数据这类事物，宾语多是研究者、从业者这类人群；例句如：The striking social phenomenon intrigues numerous sociologists. 这一引人注目的社会现象激起了众多社会学家的研究兴趣。conservationist指自然资源保护工作者。连词for表原因，书面正式，多用于议论文，放在主句之后补充理由，区别于because，for偏向对现象做事后解释，语气弱一些，不用于回答why提问。a good insurance against extinction属于很出彩的比喻写法，insurance本指保险，这里引申为“抵御……的保障”，insurance against + 不利名词是固定搭配，against后面接想要规避的风险，这里风险是extinction（物种灭绝）。例句如：Sound psychological resilience is a good insurance against life frustrations. 良好的心理韧性是抵御人生挫折的有力保障。

习语in spades意为“大量地、程度很高”，放在句末修饰动词have，用来强化数量、程度，口语化但在外刊中高频出现，写作时可以拿来替换普通的very、a great deal。例句如：Young generations have creativity and imagination in spades. 年轻一代拥有极为丰富的创造力与想象力。

句式One question that needed an answer was whether...是议论文很实用的表达，写作里用来引出亟待验证的研究猜想，相当于“有一个亟待解答的问题是……”，适合阅读摘记，也可以仿写用于实验、话题探讨类段落，用来抛出学界存在的一种假想，后文再给出证据推翻或者证实该猜想。例句如：One question that needed an answer was whether remote working would permanently reshape people's office-going habits. 有一个亟待解答的问题是，远程办公是否会永久改变人们的通勤上班习惯。One question that needed an answer was whether excessive online information would weaken individuals' ability of independent thinking. 有一个亟待解答的问题是，海量网络信息是否会削弱人们的独立思考能力。

短语taken a wrong turn本义是走错路，这里拟人化描述鲑鱼洄游路线发生偏移，不局限于生物语境，写作中也可以引申指人或群体做出错误抉择。例句如：Without rational guidance, some young people may take a wrong turn in pursuit of quick material gains. 缺少理性引导，部分年轻人在追逐物质速成的过程中可能会走上歧途。Faced with complicated social challenges, a government may take a wrong turn if it ignores public real-life demands. 面对复杂社会挑战，如果忽视民众现实诉求，政府有可能做出方向性错误。introduce...into...此处不是简单的“介绍”，熟词僻义，表示“将...引入、传入”，文中指基因的流入，在科技类文本经常用来描述物种、技术、文化传入某个区域。例句如：Global trade and cultural exchanges introduce diverse values into local communities. 全球贸易与文化交流把多元价值观带入本土社群。

从行文逻辑来看，作者先摆出一种流行的假想解释，再在下一句用however转折否定该猜想，先立假说再驳假说，这是科学文章常见的论证套路：先把外界可能会想到的解释摆出来，再用实验证据予以排除，以此凸显自身研究结论的合理性，避免读者产生相关疑问。模板句式：One question that needed an answer was whether... ..., however,... 例句：One question that needed an answer was whether economic growth alone could solve all social problems. Economic data, however, bear no close relation to people's sense of happiness. 有一个亟待解答的问题是：单凭经济增长能否解决全部社会问题。然而，经济数据和民众幸福感并没有密切关联。注意这里的bear no close relation to一并复用，考研写作辩证段落可用，表示“与……并无密切关联”，替代简单版have nothing to do with，提升表达档次。例句：High test scores bear no close relation to one's practical problem-solving competence. 考试高分和一个人实际解决问题的能力并没有密切关联。